Bell's Palsy

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KEYWORDS

• Bell's palsy • Bell palsy • Idiopathic facial nerve palsy • Peripheral facial palsy

KEY POINTS

- Bell's palsy is a common, idiopathic condition causing acute onset of unilateral facial paralysis in the distribution of peripheral cranial nerve VII.
- Diagnose Bell's palsy based on a careful history and physical examination.
- Treat all adult patients with Bell's palsy with oral corticosteroids to increase rates of complete recovery.
- Treat all adult patients with Bell's palsy with antiviral agents to reduce rates of synkinesis.
- Refer patients with incomplete recovery to physical therapy and to an appropriate specialist.

INTRODUCTION

Bell's palsy, also referred to as idiopathic facial paralysis, is an acute weakness or paralysis of the facial muscles related to swelling of the facial nerve and has no apparent cause. Primary care physicians play a pivotal in the clinical evaluation, diagnosis, treatment, and follow-up of patients with Bell's palsy, and can manage most patients with Bell's palsy in the primary care setting. This review will provide an evidence-based approach to the diagnosis and management of Bell's palsy.

EPIDEMIOLOGY

The estimated annual incidence of Bell's palsy is 20 to 30 per 100,000 people. Although the highest incidence occurs in individuals between 15 and 45 years of age, it can occur at any age.[1–4] Women and men are equally affected,[1,2] and both sides of the face are equally affected.[3,5] Conditions associated with increased risk of Bell's Palsy include diabetes,[2,6] pregnancy,[1,7] hypertension,[2] immunosuppression,[2] influenza A, and other upper respiratory illnesses.[3,5]

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PATHOGENESIS

While Bell's palsy has no apparent cause, the pathophysiology is associated with nerve edema and subsequent mechanical compression of the facial nerve, leading to weakness or paralysis in the distribution of the facial nerve (Fig. 1). Bell's palsy may also affect the sensory and autonomic function of the facial nerve. The facial nerve carries sensation from the external auditory canal, pinna, mastoid, and the palatal mucosa. Impaired function of the chorda tympani branch can affect taste and cause abnormal salivation. Damage to the petrosal nerve branch, which innervates the lacrimal glands, leads to decreased lacrimation. As fibers from the facial nerve also innervate the stapedius, some patients may have hyperacusis.[8,9] Potential etiologies include ischemic neuritis or reactivation of Herpes Simplex Virus Type 1 in the geniculate ganglion.[8]

CORONAVIRUS DISEASE 2019 AND BELL'S PALSY

With the onset of the coronavirus disease 2019 (COVID-19) pandemic, questions arose regarding the relationship between Bell's palsy, COVID-19 infection, and vaccinations against COVID-19. A systematic review and meta-analyses analyzed

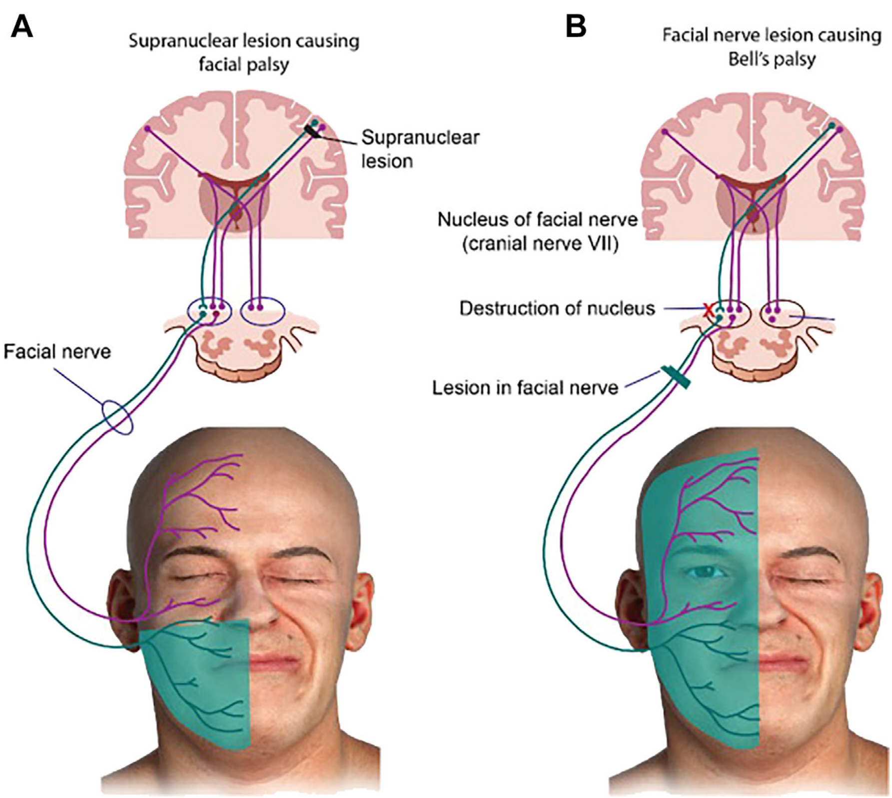

Fig. 1. Etiology of facial palsy. (*A*) Supranuclear lesion causing upper motor nerve palsy. (*B*) Facial nerve lesion causing lower motor nerve palsy, consistent with Bell's palsy. (S. M. Balaji, Padma Preetha Balaji, Chapter 40 - Sensory and motor disturbances of the orofacial region, Textbook of Oral and Maxillofacial Surgery, 4th Edition. Elsevier India, 2022.)

randomized controlled trial data and observational data. There was an increased incidence of Bell's palsy among individuals receiving either a messenger-RNA (m-RNA) vaccine or a viral vector vaccine in randomized placebo controlled-trials of both vaccine types (odds ratio [OR] 3.0, 95% confidence of interval [CI], 1.1–8.18).[10] A pooled analysis of observational studies of m-RNA vaccines, which included 13,518,026 observed doses and 13,510,701 unvaccinated individuals showed no association between vaccine administration and Bell's palsy (OR 0.7, 95% CI, 0.42–1.16).[9] This data suggest an association between COVID vaccination and incident Bell's palsy, but does not prove a causal relationship. Furthermore, Bell's palsy was observed to be more common following infection with COVID-19 than following vaccination (OR 3.23; 95% CI, 1.57–6.62).[10]

CLINICAL PRESENTATION

Bell's palsy presents acutely with unilateral facial weakness or paralysis involving the forehead developing over 24 to 48 h.[4,8] Rarely, Bell's palsy can be bilateral. The palsy most noticeably affects the motor function of the facial nerve but affects the autonomic and sensory function as well.[8] Patients may complain of alteration in taste, sensation, or hearing on the affected side.[8] The history and physical examination are essential to the diagnosis of Bell's palsy and can reliably lead to the diagnosis, as well as identify findings that raise concern for an alternative and potentially more serious diagnosis. The examiner should find no other neurologic abnormalities in addition to findings consistent with peripheral facial nerve involvement.[4,8]

EXAMINATION

Bell's palsy presents with drooping of the mouth, flattening of the nasolabial fold, inability to close the affected eye, and smoothing of the brow on the affected side.[8,9] (see **Figs. 1** and **2**) The provider should perform complete neurologic and otolaryngologic examinations with the primary goal of differentiating between an upper, or central, and lower, or peripheral, motor neuron lesion affecting the facial nerve.[11–13] By

Fig. 2. Peripheral facial nerve palsy on the right side. Typical presentation of peripheral paralysis of the facial nerve (cranial nerve VII) on the right side. (*A*) Face at rest. The skin creases in the right half of the face have smoothed out. (*B*) When the patient attempts to raise his eyebrows, only the left half of the forehead displays wrinkles, due to loss of function of right frontalis muscle. (*C*) When the patient attempts to shut both eyes, he is unable to do so on the right side (lagopthhalmos) due to loss of function of the right orbicularis oculi muscle. (*D*) When patient attempts to wrinkle his nose, he is unable to do so on the right side of face. (*E*) When patient tries to whistle, no tone is produced but air escapes through the lips on the paralyzed side, due to loss of function of the orbicularis oris muscle. (Paulsen, Waschke, Sobotta Atlas of Human Anatomy, 17th Edition. 2023 © Elsevier GmbH, Urban & Fischer, Munich.)

definition, Bell's palsy is a peripheral lesion, which causes weakness of all the muscles involved in facial expression on the affected side. This includes the frontalis muscle which raises the eyebrow; orbicularis oculi muscle which closes the eyelids and is involved in tear production and the corneal reflex; the orbicularis oris muscle which closes and protracts the lips, is involved with speech, and is essential to swallowing, chewing, and sucking; the buccinator muscle which pulls the corners of the mouth laterally; and the platysma muscle which pulls down the mandible and pulls the corners of the lips out to the side and down.[8,9]

If the patient can raise their eyebrow on the affected side, suggesting that the frontalis muscle is not affected, consider a central lesion, particularly a stroke. Additionally, consider an alternative diagnosis if the initial examination reveals any neurologic findings in addition to those associated with the facial muscles described earlier as these findings would indicate a lesion affecting structures other than just cranial nerve VII.[8,9] (see **Fig. 1**) If the patient presents with ipsilateral recurrent Bell's palsy,[14] symptoms continue to progress beyond the initial 72 h after onset, the symptoms fluctuate, or there is bilateral involvement at initial presentation, additional evaluation is required.[15]

While Bell's palsy is defined as idiopathic facial nerve palsy, there are many conditions that cause peripheral facial nerve palsy or central processes that cause facial weakness. Findings that suggest an alternative diagnosis include gradual symptom onset, recent tick or vaccine exposure, tinnitus, hearing loss or vertigo, signs of infection, head and neck cancer, and limb or bulbar weakness.[8,9] Central causes will typically spare the forehead. Gradual onset and mental status changes suggest brain tumor, either primary, or if a history of cancer, metastatic. Abrupt onset associated with ipsilateral extremity weakness or other neurologic symptoms indicate a stroke. Multiple sclerosis causing facial weakness is also associated with other neurologic symptoms. Viral diseases that cause peripheral nerve palsy include COVID-19, cytomegalovirus, Epstein-Barr virus, herpes simplex, human immunodeficiency virus (HIV), influenza, mumps, and rubella—weakness associated with these infections will typically be preceded or accompanied by the associated viral syndrome. Ramsay Hunt syndrome, due to herpes zoster, is characterized by prodromal pain and a vesicular eruption in the ear canal and/or palate. Otitis media can cause peripheral nerve palsy of gradual onset, associated with pain, fever, and conductive hearing loss. Suspect Lyme disease if there is a history of tick exposure, residence in or travel to endemic areas, rash and/or arthralgias or bilateral facial weakness. Autoimmune or granulomatous conditions such as sarcoidosis, myasthenia gravis, or Guillain-Barre syndrome are more often bilateral at presentation. Cholesteatoma and head and neck cancers will typically have a gradual onset and associated findings on physical examination. Weakness associated with tinnitus and hearing loss suggests the diagnosis of acoustic neuroma.[4,8,9]

DIAGNOSTIC TESTING

Laboratory testing is not required to diagnose Bell's palsy. The provider should consider laboratory testing if the patient's history suggests an identifiable cause of peripheral facial nerve palsy, such as Lyme disease, HIV, or sarcoidosis. One guideline suggests obtaining a complete blood count with differential as an increased neutrophil-to-lymphocyte ratio has been associated with poor prognosis. However, the review upon which this recommendation is based demonstrated evidence of publication bias and was unable to provide a clear cut-off separating patients with poor prognosis from those with good prognosis.[13,16]

Imaging is not required to diagnose Bell's palsy if the physical examination findings are consistent with the diagnosis. There is a small risk of lacunar infarct affecting the lower pons that presents as a peripheral palsy; this is exceedingly rare.[17] Imaging is indicated if the patient presents with recurrent ipsilateral Bell's palsy, symptom progression beyond the initial 72 h following onset, fluctuate, or are bilateral. MRI head or MRI orbits/face/neck with and without intravenous contrast are the 2 recommended studies to evaluate for suspected tumor.[18]

CLINICAL ASSESSMENT TOOLS

Patients can experience a spectrum of symptom severity. Two grading scales are commonly used to classify severity, the House-Brackmann (**Fig. 3**) and Sunnybrook (**Fig. 4**) scales. Classifying severity can help provide appropriate anticipatory guidance, and aid in treatment recommendations, particularly early physical therapy for individuals with scores indicating higher severity.

PROGNOSIS

Most patients with Bell's palsy will recover completely but long-term sequelae can occur. Complete spontaneous recovery occurs in up to 85% of patients experiencing Bell's palsy within 3 weeks, with at least partial recovery in the following 3 to 6 months.[19] Regarding special populations, up to 90% of pregnant patients and children under the age of 14 will experience complete spontaneous recovery.[7,20] Clinically useful indicators of poor prognosis for complete spontaneous recovery include bilateral involvement and more severe symptoms (as determined by the Sunnybrook or House-Brackman assessments).[7,16,21] The recurrence rate of Bell's palsy has been estimated at 6.5% with a mean interval of 10 years. Of patients who experience a recurrence of Bell's palsy, approximately two-thirds subsequently achieve complete spontaneous recovery.[14]

Long-term sequelae of Bell's palsy, especially the impairment of smiling, may trigger anxiety, depression, and other severe psychologic problems.[22] Synkinesis, which is

HOUSE-BRACKMANN FACIAL NERVE GRADING SYSTEM

Grade I: Normal function

- Normal facial function in all areas

Grade II: Mild dysfunction

- Slight weakness noticeable on close inspection; may have very slight synkinesis

Grade III: Moderate dysfunction

- Obvious weakness or disfiguring asymmetry; normal symmetry and tone at rest; incomplete eye closure

Grade IV: Moderately severe dysfunction

- Obvious weakness or disfiguring asymmetry; normal symmetry and tone at rest; incomplete eye closure

Grade V: Severe dysfunction

- Only barely perceptible motion; asymmetry at rest

Grade VI: Total paralysis

- No movement

Fig. 3. House-Brackmann facial nerve grading scale. (Edward T. Bope, Rick D. Kellerman, Conn's Current Therapy 2012, 1 edition. Saunders, 2012.)

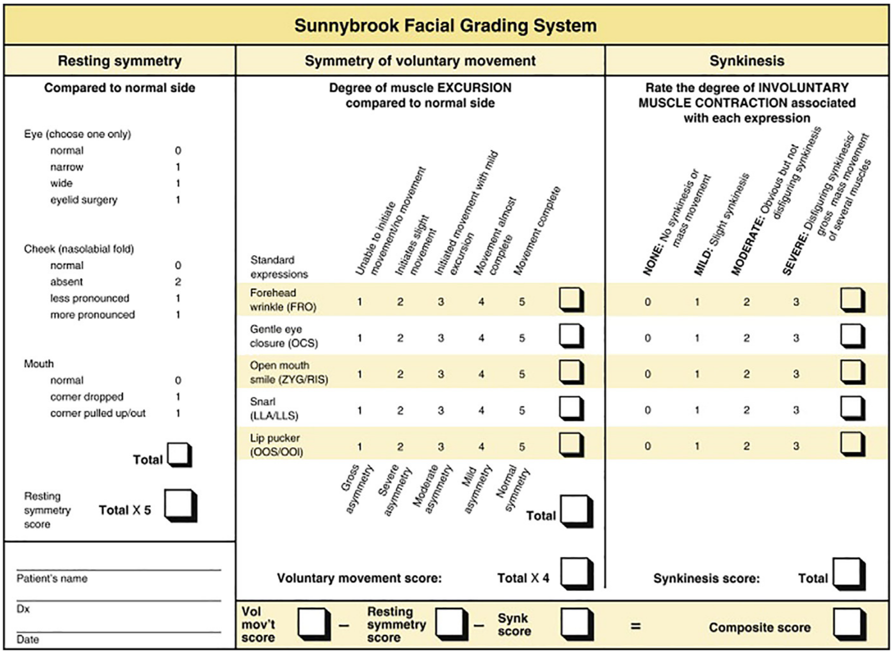

Fig. 4. Sunnybrook grading scale. (Helen Hartley, Wendy Blumenow, Rebecca Williams, Adel Fattah, 7 - Assessment and Grading of Synkinesisand Facial Palsy, Editor(s): Babak Azizzadeh, Charles Nduka, Management of Post-Facial Paralysis Synkinesis, Elsevier, 2022.)

caused by misdirection of regenerating nerve fibers, can have motor and autonomic manifestations in patients with Bell's palsy. Motor symptoms include involuntary blinking when smiling and involuntary lip movement with blinking. Autonomic symptoms can include gustatory eye watering (crocodile tears) and gustatory sweating. When severe, synkinesis can be associated with facial muscle spasm.[8] Synkinesis can affect up to 26% of patients 1 year after Bell' Palsy onset.[3]

CURRENT CONTROVERSIES

No guidelines currently recommend surgery, although all note that there is disagreement among experts regarding this recommendation.[11–13] There are 2 approaches to surgical decompression of the facial nerve, the transmastoid decompression (TMD) approach and the middle fossa decompression (MFD) approach. The theoretic rationale for surgery is that it reduces pressure, and thus ischemia, on the facial nerve in cases of severe or complete paralysis, which will lead to improved recovery. Three recent systematic reviews provide conflicting conclusions regarding the effectiveness of surgical approaches. A 2020 Cochrane review of 2 controlled trials (1 randomized, 1 quasi-randomized) showed no improvement in long-term outcomes of patients undergoing surgery compared to those having standard medical therapy.[23] A 2019 systematic review and meta-analysis, which included 5 studies excluded from the Cochrane review for methodologic flaws, showed that surgery had a positive effect.[24] Another systematic review of 7 cohort studies found that the TMD approach was no better than standard medical therapy, but that early surgery via the MFD approach (within 14 d of onset) was associated with improved outcomes.[25] When pooling all results, no improvement for surgical intervention was found.[25] Potential adverse effects of surgery include hearing loss, tinnitus, and cerebrospinal fluid leaks, depending upon the approach.[25] Only consider surgical referral for patients with complete paralysis or severe Bell's palsy who have not responded to medical therapy. Refer these patients to advanced centers with expertise in facial nerve decompression surgery.[12]

Another area of controversy in treatment of Bell's palsy is the role of acupuncture. None of the current guidelines recommend acupuncture.[11–13] A Cochrane review found no evidence that acupuncture was beneficial.[26] A systematic review and meta-analysis found that acupuncture did show a slight benefit when added to standard therapy, but the quality of the included trials was poor and the authors did not find that the available evidence supported the use of acupuncture.[27] Well done randomized controlled trials are needed before acupuncture can be routinely offered as a safe and effective treatment modality.

MEDICAL/PHARMACOLOGIC MANAGEMENT

All patients 16 year of age and older with Bell's palsy should be treated with corticosteroids, within 72 h of onset.[11–13] Multiple randomized control trials and meta-analyses have confirmed their effectiveness in improving rates of complete recovery, with a Number Needed to Treat (NNT) = 10.[3,28,29] Less strong evidence, based on indirect comparison, suggests that a total dose of more than 450 mg for a course of steroids improves recovery rates compared to cumulative doses of 450 mg or less.[28] A typical dose studied in trials is 1 mg/kg per day of prednisone for 5 d, followed by a 5 d taper.[3,28] The French Guidelines recommend very high doses of steroids (120–200 mg/d) in severe cases.[13] A systematic review of retrospective cohort studies examining high dose versus standard dose therapy did not find improvement in outcomes (OR 0.58, 95% CI 0.27–1.22) except in those patients with severe Bell's palsy who did not receive antiviral medication (OR 0.14, 95% CI 0.04–0.54).[30] Offering patients with severe Bell's palsy who are unable to take antiviral medications a high dose of oral corticosteroids is an option for shared decision-making.

While there are limited data to guide the treatment of pregnant women with Bell's palsy, expert consensus is to treat pregnant women with oral corticosteroids.[7]

Do not offer prednisone to children. A large, randomized trial comparing oral corticosteroids to placebo among children under age 18 presenting to an emergency department showed there was no difference in recovery rates at 3 mo, 6 mo, or at 1 y, respectively, and no serious adverse events were reported.[31,32] There was no difference in response rates between children below age 12 and those 12 and older.[32] Earlier studies of oral corticosteroids and antiviral agents in adults did include adolescents aged 16 and older.[3,28,29] These disparate data should be discussed with patients and their parent or guardian to achieve a shared decision for adolescents between the ages of 16 and 18.

Offer antiviral agents to adults with Bell's palsy, in addition to corticosteroids, to reduce the risk of synkinesis. A Cochrane review provides high quality evidence that adding antiviral agents to steroids reduces the risk of synkinesis, NNT = 12.[5] Meta-analysis and network meta-analyses (NMA) provide weak evidence that combined therapy with steroids and oral antivirals reduce rates of incomplete recovery and improve patient satisfaction with recovery.[5,29] One NMA identified famciclovir as the most effective agent to combine with a steroid to improve rates of recovery.[33] This same NMA suggested that acyclovir was most effective at reducing rates of synkinesis.[33] Antivirals alone provide no benefit to patients with Bell's palsy and they should not be prescribed without corticosteroids.[11–13,34] Currently, the available evidence does not strongly support the use of one antiviral agent over another. Prescribe acyclovir, famciclovir, or valacyclovir. Dosages in studies range from 1200 mg to 2400 mg/d in divided doses for 7 to 10 d for acyclovir, 250 mg TID for 7 d for famciclovir and 500 mg BID for 5 d to 1 gm TID for 7 d for valacyclovir.

All guidelines recommend, on the strength of expert opinion, protective eye care to patients with Bell's palsy who are unable to close their eye. This consists of application of topical moisturizers, and taping the eyelid closed at bedtime, to prevent corneal damage.[11–13]

Refer patients who do not experience complete recovery to physical therapy. A 2011 Cochrane review and a more recent systematic review of randomized or quasi-randomized trials provide weak evidence that physical therapy can help patients achieve full recovery or improvement.[35,36] The available evidence does not allow for precise recommendations regarding the timing of physical therapy referral or the physical therapy regimen.

Patients with synkinesis or persistent weakness may also benefit from treatment with botulinum toxin. One review showed improvement in quality-of-life measures following treatment with Botox.[37] Another systematic review showed that Botox improves outcomes in patient with synkinesis.[38] Due to the diversity of individual patient presentations, it is not possible to recommend specific treatment regimens with Botox, in terms of type of toxin, dosage, injection site, or frequency of repetition.[38]

Available data do not support a role for hyperbaric oxygen, electrical stimulation, laser therapy, stellate ganglion blocks, or intratympanic steroid injections in the treatment of Bell's palsy.[11–13,39–41]

PATIENT MONITORING

Patients with BP should be seen periodically to assess for resolution of facial weakness and any complications, such as corneal ulceration. Refer patients to an appropriate specialist if new or worsening weakness develops, complete or severe paralysis does not improve within 1 to 2 w, eye symptoms develop, or if there is incomplete recovery at 3 mo.[11–13]

SUMMARY

Bell's palsy is characterized by abrupt onset of unilateral facial weakness and can be associated with alterations in taste or sensation, as well as hyperacusis. Laboratory testing or imaging is not needed for patients with a typical presentation. Gradual onset, sparing of the forehead, pain, and other neurologic or systemic symptoms or examination findings suggest and alternative diagnosis. The prognosis for complete recovery in Bell's palsy is good. Treat all adult patients with oral corticosteroids to increase rates of complete recovery and with oral antivirals to reduce rates of synkinesis. Refer patients with severe Bell's palsy or incomplete recovery to physical therapy, and if patients do not improve with treatment refer to an appropriate specialist.

CLINICS CARE POINTS

- Diagnose Bell's palsy based on a careful history and physical.
- Provide eye protection to all patients with Bell's palsy who cannot completely close their eye.
- Treat all adult patients with Bell's palsy with oral corticosteroids at a dose of 1 mg/kg for 5 d, followed by a 5-d taper.

- Treat all adult patients with oral acyclovir, famciclovir, or valacyclovir to reduce rates of synkinesis.
- Do not prescribe steroids or antivirals to children with Bell's palsy.

DISCLOSURE

The authors have nothing to disclose.

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